EX.NO: 1 **Find the Second Largest Element (without sorting)**

PROGRAM:

```
import java.util.Scanner;
public class SecondLargest
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;
        for (int i = 0; i < n; i++)
        {
            if (arr[i] > largest)
            {
                secondLargest = largest;
                largest = arr[i];
            } else if (arr[i] > secondLargest && arr[i] != largest)
            {
                secondLargest = arr[i];
            }
```

```java
        }
        if (secondLargest == Integer.MIN_VALUE)
        {
            System.out.println("No second largest element found.");
        } else
        {
            System.out.println("Second Largest Element: " + secondLargest);
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter the number of elements: 7
Enter the array elements:
100
700
450
400
800
790
900
Second Largest Element: 800
```

EX.NO: 2 **Find the Second Smallest Element**

PROGRAM:

```
import java.util.Scanner;
public class SecondSmallest
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;
        for (int i = 0; i < n; i++)
        {
            if (arr[i] < smallest)
            {
                secondSmallest = smallest;
                smallest = arr[i];
            } else if (arr[i] < secondSmallest && arr[i] != smallest) {
                secondSmallest = arr[i];
            }
```

```
        }
        if (secondSmallest == Integer.MAX_VALUE)
        {
            System.out.println("No second smallest element found.");
        } else
        {
            System.out.println("Second Smallest Element: " + secondSmallest);
        }
        sc.close();
    }
}
```

OUTPUT:

```
Enter the number of elements: 5
Enter the array elements:
55
11
88
77
60
Second Smallest Element: 55
```

EX.NO: 3 **Move All Zeros to the End**

PROGRAM:

```
import java.util.Scanner;
public class MoveZerosToEnd
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int index = 0;
        for (int i = 0; i < n; i++)
        {
            if (arr[i] != 0)
            {
                int temp = arr[index];
                arr[index] = arr[i];
                arr[i] = temp;
                index++;
            }
        }
        System.out.println("Array after moving zeros to the end:");
```

```
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the array elements:

10

5

00

55

000000

Array after moving zeros to the end:

10 5 55 0 0

EX.NO: 4 **Rotate Array to the Left by One Position**

PROGRAM:

```
import java.util.Scanner;
public class LeftRotateByOne
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int first = arr[0];
        for (int i = 0; i < n - 1; i++)
        {
            arr[i] = arr[i + 1];
        }
        arr[n - 1] = first;
        System.out.println("Array after left rotation by one position:");
        for (int i = 0; i < n; i++)
        {
            System.out.print(arr[i] + " ");
        }
        sc.close();
```

```
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the array elements:

10

20

30

40

50

Array after left rotation by one position:

20 30 40 50 10

EX.NO: 5 **Rotate Array to the Right by One Position**

PROGRAM:

```
import java.util.Scanner;
public class RightRotateByOne
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int last = arr[n - 1];
        for (int i = n - 1; i > 0; i--)
        {
            arr[i] = arr[i - 1];
        }
        arr[0] = last;
        System.out.println("Array after right rotation by one position:");
        for (int i = 0; i < n; i++)
        {
            System.out.print(arr[i] + " ");
        }
```

```
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the array elements:

10

20

30

40

50

Array after right rotation by one position:

50 10 20 30 40

EX.NO: 6 **Find Missing Number (1 to N)**

PROGRAM:

```
import java.util.Scanner;

public class FindMissingNumber

{

   public static void main(String[] args)

   {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter the value of N: ");

      int n = sc.nextInt();

      int[] arr = new int[n - 1];

      System.out.println("Enter " + (n - 1) + " elements:");

      for (int i = 0; i < n - 1; i++) {

         arr[i] = sc.nextInt();

      }

      int totalSum = n * (n + 1) / 2;

      int arraySum = 0;

      for (int i = 0; i < n - 1; i++) {

         arraySum += arr[i];

  }

     int missing = totalSum - arraySum;

     System.out.println("Missing Number: " + missing);

      sc.close();

   }

}
```

OUTPUT:

Enter the value of N: 6

Enter 5 elements:

1

2

3

4

6

Missing Number: 5

EX.NO: 7 **Find the Element that Appears Only Once**

PROGRAM:

```
import java.util.Scanner;
public class SingleElement
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int result = 0;
        for (int i = 0; i < n; i++)
        {
            result ^= arr[i];
        }
        System.out.println("Element that appears only once: " + result);
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the array elements:

23

32

25

52

55

Element that appears only once: 45

EX.NO: 8 **Find the First Repeating Element**

PROGRAM:

```
import java.util.Scanner;
public class FirstRepeatingElement
{
  public static void main(String[] args)
  {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the number of elements: ");
    int n = sc.nextInt();
    int[] arr = new int[n];
    System.out.println("Enter the array elements:");
    for (int i = 0; i < n; i++)
    {
      arr[i] = sc.nextInt();
    }
    boolean found = false;
    for (int i = 0; i < n; i++)
    {
      for (int j = i + 1; j < n; j++)
      {
        if (arr[i] == arr[j]) {
          System.out.println("First Repeating Element: " + arr[i]);
          found = true;
          break;
        }
      }
      if (found)
```

```
            {
                break;
            }
        }
        if (!found)
        {
            System.out.println("No repeating element found.");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the array elements:

66

66

77

52

25

First Repeating Element: 66

EX.NO: 9 **Find the First Non-Repeating Element**

PROGRAM:

```
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;
public class FirstNonRepeatingElement
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        Map<Integer, Integer> map = new HashMap<>();
        for (int i = 0; i < n; i++)
        {
            map.put(arr[i], map.getOrDefault(arr[i], 0) + 1);
        }
        int result = -1;
        for (int i = 0; i < n; i++)
        {
            if (map.get(arr[i]) == 1)
            {
                result = arr[i];
```

```
            break;
          }
        }
        if (result != -1)
        {
          System.out.println("First non-repeating element: " + result);
        } else
        {
          System.out.println("No non-repeating element found");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 6

Enter the elements:

2

2

44

5

6

88

First non-repeating element: 44

EX.NO: 10 **Check if Array is Sorted**

PROGRAM:

```
import java.util.Scanner;
public class CheckArraySorted
{
   public static void main(String[] args)
   {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter the number of elements: ");
      int n = sc.nextInt();
      int[] arr = new int[n];
      System.out.println("Enter the elements:");
      for (int i = 0; i < n; i++)
      {
         arr[i] = sc.nextInt();
      }
      boolean isSorted = true;
      for (int i = 0; i < n - 1; i++)
      {
         if (arr[i] > arr[i + 1])
         {
            isSorted = false;
            break;
         }
      }
      if (isSorted)
      {
         System.out.println("Array is sorted");
```

```
        } else
        {
            System.out.println("Array is not sorted");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

5

10

15

20

25

Array is sorted

EX.NO: 11 Reverse Array Without Using Another Array

PROGRAM:

```
import java.util.Scanner;
public class ReverseArray
{
  public static void main(String[] args)
  {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the number of elements: ");
    int n = sc.nextInt();
    int[] arr = new int[n];
    System.out.println("Enter the elements:");
    for (int i = 0; i < n; i++)
    {
      arr[i] = sc.nextInt();
    }
    int start = 0;
    int end = n - 1;
    while (start < end) {
      int temp = arr[start];
      arr[start] = arr[end];
      arr[end] = temp;
      start++;
      end--;
    }
    System.out.println("Reversed array:");
    for (int i = 0; i < n; i++)
    {
```

```
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

44

88

22

25

52

Reversed array:

52 25 22 88 44

EX.NO: 12 **Find Maximum Difference Between Two Elements**

PROGRAM:

```
import java.util.Scanner;
public class MaximumDifference
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int minElement = arr[0];
        int maxDifference = arr[1] - arr[0];
        for (int i = 1; i < n; i++) {
            int difference = arr[i] - minElement;
            if (difference > maxDifference) {
                maxDifference = difference;
            }
            if (arr[i] < minElement) {
                minElement = arr[i];
            }
        }
        System.out.println("Maximum difference: " + maxDifference);
        sc.close();
```

```
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

45

54

55

78

87

Maximum difference: 42

EX.NO: 13 **. Leaders in an Array**

PROGRAM:

```
import java.util.Scanner;
public class LeadersInArray
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Leaders in the array:");
        int maxFromRight = arr[n - 1];
        System.out.print(maxFromRight + " ");
        for (int i = n - 2; i >= 0; i--) {
            if (arr[i] >= maxFromRight) {
                maxFromRight = arr[i];
                System.out.print(maxFromRight + " ");
            }
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

85

58

66

33

22

Leaders in the array:

22 33 66 85

EX.NO: 14 **Find Equilibrium Index**

PROGRAM:

```
import java.util.Scanner;
public class EquilibriumIndex {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int totalSum = 0;
        for (int i = 0; i < n; i++) {
            totalSum += arr[i];
        }
        int leftSum = 0;
        int equilibriumIndex = -1;
        for (int i = 0; i < n; i++) {
            totalSum -= arr[i];
            if (leftSum == totalSum) {
                equilibriumIndex = i;
                break;
            }
            leftSum += arr[i];
        }
```

```java
        if (equilibriumIndex != -1) {
            System.out.println("Equilibrium index: " + equilibriumIndex);
        } else {
            System.out.println("No equilibrium index found");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

1

3

5

2

2

Equilibrium index: 2

EX.NO: 15 **Find Pair with Given Sum**

PROGRAM:

```
import java.util.HashSet;
import java.util.Scanner;
public class PairWithGivenSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter the target sum: ");
        int target = sc.nextInt();
        HashSet<Integer> set = new HashSet<>();
        boolean found = false;
        for (int i = 0; i < n; i++) {
            int required = target - arr[i];
            if (set.contains(required)) {
                System.out.println("Pair found: " + required + " + " + arr[i] + " = " + target);
                found = true;
                break;
            }
            set.add(arr[i]);
        }
```

```
        if (!found) {
            System.out.println("No pair found with the given sum");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

2

7

11

15

5

Enter the target sum: 20

Pair found: 15 + 5 = 20

EX.NO: 16 **Find Majority Element**

PROGRAM:

```
import java.util.Scanner;
public class MajorityElement
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int candidate = arr[0];
        int count = 1;
        for (int i = 1; i < n; i++)
        {
            if (arr[i] == candidate)
            {
                count++;
            } else
            {
                count--;
                if (count == 0)
                {
```

```
            candidate = arr[i];
            count = 1;
          }
        }
      }
      count = 0;
      for (int i = 0; i < n; i++)
      {
        if (arr[i] == candidate)
        {
          count++;
        }
      }
      if (count > n / 2)
      {
        System.out.println("Majority element: " + candidate);
      } else
      {
        System.out.println("No majority element found");
      }
      sc.close();
    }
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

2

2

1

2

3

Majority element: 2

EX.NO: 17 **Rearrange Array in Wave Form**

PROGRAM:

```
import java.util.Scanner;
public class WaveArray
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }

        for (int i = 0; i < n; i += 2)
        {
            if (i > 0 && arr[i] < arr[i - 1])
            {
                int temp = arr[i];
                arr[i] = arr[i - 1];
                arr[i - 1] = temp;
            }
            if (i < n - 1 && arr[i] < arr[i + 1])
            {
                int temp = arr[i];
```

```
            arr[i] = arr[i + 1];
            arr[i + 1] = temp;
          }
        }
        System.out.println("Array in Wave Form:");
        for (int i = 0; i < n; i++)
        {
          System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 5

Enter the elements:

5

2

2

5

1

Array in Wave Form:

5 2 5 1 2

EX.NO: 18 **Find Longest Consecutive Increasing Sequence**

PROGRAM:

```
import java.util.Scanner;
public class LongestIncreasingSequence
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enterthe array element:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int maxLength = 1;
        int currentLength = 1;
        int start = 0;
        int maxStart = 0;
        for (int i = 1; i < n; i++)
        {
            if (arr[i] > arr[i - 1])
            {
                currentLength++;
            } else
            {
                currentLength = 1;
```

```
            start = i;
        }
        if (currentLength > maxLength)
        {
            maxLength = currentLength;
            maxStart = start;
        }
    }
    System.out.println("Longest Consecutive Increasing Sequence:");
    for (int i = maxStart; i < maxStart + maxLength; i++)
    {
        System.out.print(arr[i] + " ");
    }
    System.out.println("\nLength = " + maxLength);
    sc.close();
  }
}
```

OUTPUT:

Enter the number of elements: 8

Enterthe array element:

1

2

3

1

2

3

4

5

Longest Consecutive Increasing Sequence:

1 2 3 4 5

Length = 5

EX.NO: 19 **Find Frequency of Every Element Without Using HashMap**

PROGRAM:

```
import java.util.Scanner;
public class FrequencyWithoutHashMap
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        boolean[] visited = new boolean[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        System.out.println("Element\tFrequency");
        for (int i = 0; i < n; i++)
        {
            if (visited[i])
                continue;
            int count = 1;
            for (int j = i + 1; j < n; j++)
            {
                if (arr[i] == arr[j])
                {
                    count++;
```

```java
                    visited[j] = true;
                }
            }
            System.out.println(arr[i] + "\t\t" + count);
        }
        sc.close();
    }
}
```

OUTPUT:

```
Enter the number of elements: 8
Enter the array elements:
2
3
2
5
3
2
6
5
Element         Frequency
2               3
3               2
5               2
6               1
```

EX.NO: 20 **Find the Smallest Positive Missing Number**

PROGRAM:

```
import java.util.Scanner;
public class SmallestPositiveMissing
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        boolean[] present = new boolean[n + 2];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
            if (arr[i] > 0 && arr[i] <= n + 1)
            {
                present[arr[i]] = true;
            }
        }
        for (int i = 1; i <= n + 1; i++)
        {
            if (!present[i]) {
                System.out.println("Smallest Positive Missing Number = " + i);
                break;
            }
        }
```

```
        sc.close();
    }
}
```

OUTPUT:

Enter the number of elements: 4

Enter the array elements:

3

4

-1

1

Smallest Positive Missing Number = 2